



WINNER RANKINGS

- | | | |
|---|---|------------|
| 1 | Horseshoe False Discovery Rate Control via Frequentist Thresholding Im | 65% |
| 2 | T-JEPA Tabular Masking with TF-Gene Interaction Structure Prevents Co- | 63% |
| 3 | Inverted JEPA Masking Aligns Pretext Directionality with TF-to-Gene Ca ◀ | 62% |
| 4 | GMM-Anchored JEPA Pretraining Grounds W_init in Regulatory Logic Beyon | 62% |
| 5 | Conditional Independence Masking in JEPA Targets Suppresses Co-express | 60% |

Inverted JEPA Masking Aligns Pretext Directionality with TF-to-Gene Causal Edges

Status: Winner · **Domain:** Self-Supervised Representation Learning / Gene Regulatory Network Inference · **Generation:** 1 · **Score:** 62%

Abstract

Gene regulatory network (GRN) inference requires distinguishing causal TF→gene regulatory edges from spurious co-expression. Current JEPA-based initializations using TF-column masking conflate TF-TF co-variation with genuine regulatory influence, because masked TF recovery is dominated by TF co-expression rather than target gene signals. We propose an inverted masking strategy: at each Stage 1 training step, k target gene columns ($k=20-30\%$ of target genes) are zeroed out, and the encoder is trained to predict their expression exclusively from the visible TF context. This alignment of pretext task directionality with the causal structure (TF→gene) ensures that W_{init} entries directly encode TF predictive power over individual target genes, not TF-TF redundancy. We further augment the objective with a self-distillation consistency term, adapted from self-distilled representation learning, enforcing stable TF-context embeddings across differently masked gene subsets. Evaluated on DREAM5 net3 (334 TFs), we predict AUROC improvement from 0.63 to >0.66 and Stage 2 AUPR gains through improved W_{init} specificity. Success is defined as statistically significant AUROC gain ($p<0.05$) over TF-column masking baseline and reduced TF-TF co-expression leakage measured by partial correlation residual analysis.

Predicting target gene expression from TF context only forces W_{init} to encode TF→gene predictive relationships, directly aligning with the causal edge directionality needed for GRN inference, whereas predicting masked TFs from target genes predominantly encodes TF co-expression.

3. BACKGROUND & LITERATURE REVIEW

- **Self-Distilled Representation Learning for Time Series** — Pieper et al. (2023)
- **Improving Few-Shot Learning with Auxiliary Self-Supervised Pretext Tasks** — Simard et al. (2021)
- **DSeq-JEPA: Discriminative Sequential Joint-Embedding Predictive Architecture** — He et al. (2025)
- **ST-GCP: a graph convolutional network model with contrastive consistency and permutation for spatial transcriptomics.** — Meng et al. (2025)
- **A Novel Graph Transformer Framework for Gene Regulatory Network Inference** — Teji et al. (2025)
- **GeSubNet: Gene Interaction Inference for Disease Subtype Network Generation** — Yang et al. (2024)
- **Transformer-based feature extraction approach for hematopoietic cancer subtype classification.** — Park et al. (2026)
- **Predicting the genetic component of gene expression using gene regulatory networks** — Mohammad et al. (2024)
- *+ 14 additional papers — see Section 8.*

4. METHODOLOGY

Modify Stage 1 JEPA: at each step, randomly select k target gene columns ($k \sim 20\text{--}30\%$), zero them out, and train the encoder to reconstruct them using only TF expression context (all TFs visible, target genes partially masked). Extract W_{init} as the learned linear mapping from TF embeddings to target gene predictions. Add a self-distillation consistency loss (momentum encoder as teacher) to stabilize TF context representations across different gene masking patterns, inspired by self-distilled time-series SSL. Evaluate on DREAM5 net3 with AUROC and Stage 2 AUPR metrics.

5. EXPECTED OUTCOMES

Predicted Impact: AUROC improvement from 0.63 to >0.66 on DREAM5 net3; Stage 2 AUPR gains of $\geq 5\%$ relative over TF-column masking baseline; reduced false positive rate attributable to TF-TF co-expression confounding.

Success Criteria: AUROC > 0.66 on DREAM5 net3 (statistically significant vs. TF-column baseline, $p < 0.05$); partial correlation residual analysis confirms reduced TF-TF co-expression signal in W_{init} ; Stage 2 AUPR improves by $\geq 5\%$ relative.

8. EVIDENCE SUMMARY

DIMENSION	SCORE	WEIGHT
EVIDENCE GROUNDING	50%	35%
Feasibility	70%	30%
Impact Potential	68%	25%
Novelty	62%	10%
Composite	62%	weighted

Why this won: Paper [\[4\]](#) (Self-Distilled Representation Learning for Time Series) provides the self-distillation consistency framework applicable to structured tabular gene expression data with correlated samples; Paper [\[0\]](#) (Improving Few-Shot Learning with Auxiliary Self-Supervised Pretext Tasks) supports the use of auxiliary SSL pretext tasks to shape representation quality toward downstream task structure. The inverted masking directly addresses the critic’s core challenge by reversing directionality: W_{init} now maps TF context → gene prediction, matching causal edge orientation.

- [1] Pieper, F., Ditschuneit, K., Genzel, M., Lindt, A., & Otterbach, J. (2023). *Self-Distilled Representation Learning for Time Series*.
- [2] Simard, N., & Lagrange, G. (2021). *Improving Few-Shot Learning with Auxiliary Self-Supervised Pretext Tasks*.
- [3] He, X., Sakai, S., Chandhok, S., Beery, S., & Yuan, K. (2025). *DSeq-JEPA: Discriminative Sequential Joint-Embedding Predictive Architecture*.
- [4] Meng, Y., Wang, Y., Guo, C., Tang, X., & Zhang, Z. (2025). *ST-GCP: a graph convolutional network model with contrastive consistency and permutation for spatial transcriptomics..*
- [5] Teji, B., & Roy, S. (2025). *A Novel Graph Transformer Framework for Gene Regulatory Network Inference*.
- [6] Yang, Z., Chen, Z., Liu, X., Kotoge, R., & Chen, P. (2024). *GeSubNet: Gene Interaction Inference for Disease Subtype Network Generation*.
- [7] Park, K. H., Lee, Y., Ding, W., Ryu, K. S., & Ryu, K. H. (2026). *Transformer-based feature extraction approach for hematopoietic cancer subtype classification..*
- [8] Mohammad, G. I., & Michael, T. (2024). *Predicting the genetic component of gene expression using gene regulatory networks*.
- [9] Liu, Y., Zhang, Z., Qiu, M., Wang, S., & Salim, F. D. (2025). *GatorSC: Multi-Scale Cell and Gene Graphs with Mixture-of-Experts Fusion for Single-Cell Transcriptomics..*
- [10] Bucur, I. G., Claassen, T., & Heskes, T. (2019). *Large-Scale Local Causal Inference of Gene Regulatory Relationships*.
- [11] Bucur, I. G., Bussel, T. v., Claassen, T., & Heskes, T. (2018). *A Bayesian Approach for Inferring Local Causal Structure in Gene Regulatory Networks*.
- [12] Thimonier, H., Costa, J. L. D. M., Popineau, F., Rimmel, A., & Doan, B. (2024). *T-JEPA: Augmentation-Free Self-Supervised Learning for Tabular Data*.
- [13] Herbach, U. (2021). *Gene regulatory network inference from single-cell data using a self-consistent proteomic field*.
- [14] Peng, W., Ma, Y., Li, C., Dai, W., & Fu, X. (2025). *Fusion of brain imaging genetic data for alzheimer's disease diagnosis and causal factors identification using multi-stream attention mechanisms and graph convolutional networks..*
- [15] Gögl, M., & Yau, C. (2026). *Var-JEPA: A Variational Formulation of the Joint-Embedding Predictive Architecture -- Bridging Predictive and Generative Self-Supervised Learning*.
- [16] Lee, Y., Willette, J. R., Kim, J., & Hwang, S. J. (2024). *Visualizing the loss landscape of Self-supervised Vision Transformer*.
- [17] Herbach, U. (2023). *Harissa: stochastic simulation and inference of gene regulatory networks based on transcriptional bursting*.
- [18] Du, Y., Xia, X., Ni, Z., Fan, S., & He, J. (2025). *Genetic Evidence Prioritizes Neurocognitive Decline as a Causal Driver of Sleep Disturbances: A Multi-Omics Analysis Identifying Causal Genes and Therapeutic Targets..*
- [19] Fakhry, A., Khafagy, R., & Ludl, A. (2023). *GENER: A Parallel Layer Deep Learning Network To Detect Gene-Gene Interactions From Gene Expression Data*.
- [20] Fang, Y., Qian, J., Wang, X., Cooper, L. A., & Zhou, B. (2026). *Sparser2Sparse: Single-shot Sparser-to-Sparse Learning for Spatial Transcriptomics Imputation with Natural Image Co-learning..*
- [21] Peng, R., Lu, Y., Sun, Q., Lu, Y., & Zhang, C. (2025). *KINDLE: Knowledge-Guided Distillation for Prior-Free Gene Regulatory Network Inference*.
- [22] Feng, J., Yu, T., & Zhang, Y. (2026). *HarveST uses a heterogeneous graph learning framework to reveal spatial transcriptomics patterns..*

